

MACDILL AFB, FL, USA

WMO: 747880

Lat: 27.850N

Lon: 82.517W

Elev: 14

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 12810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 38.9	(c) 43.1	(d) 25.8	(e) 19.8	(f) 46.4	(g) 30.0	(h) 24.1	(i) 50.3	(j) 20.8	(k) 63.6	(l) 18.9	(m) 64.5	(n) 6.3	(o) 10	(p) 0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.5	(c) 93.4	(d) 79.0	(e) 92.3	(f) 78.8	(g) 91.0	(h) 78.4	(i) 82.7	(j) 88.8	(k) 81.9	(l) 88.2	(m) 81.1	(n) 87.3	(o) 7.5	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.3	162.6	85.3	80.3	157.5	84.8	79.4	152.6	84.3	46.4	89.0	45.4	88.3	44.5	87.4	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a) 19.1	(b) 17.1	(c) 15.6	(d) 15.6	(e) 33.7	(f) 95.8	(g) 4.4	(h) 1.2	(i) 30.5	(j) 96.6	(k) 27.9	(l) 97.3	(m) 25.5	(n) 98.0	(o) 22.2	(p) 98.9	
(a) 19.1	(b) 17.1	(c) 15.6	(d) 15.6	(e) 29.9	(f) 83.5	(g) 5.2	(h) 1.4	(i) 26.2	(j) 84.4	(k) 23.2	(l) 85.2	(m) 20.3	(n) 86.0	(o) 16.6	(p) 87.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.7	60.6	64.0	67.8	73.0	78.5	82.3	82.8	83.2	81.8	76.6	68.7	64.6
	DBStd	9.60	8.34	7.41	6.41	4.85	3.51	2.36	2.16	2.15	2.50	5.09	6.48	8.04
	HDD50	23	15	3	0	0	0	0	0	0	0	0	0	5
	HDD65	490	183	97	49	7	0	0	0	0	0	4	39	111
	CDD50	8679	343	396	553	690	885	968	1018	1029	955	823	562	457
	CDD65	3671	46	71	136	247	420	518	553	564	505	362	151	98
	CDH74	33831	217	351	786	1759	3876	5489	6022	6140	4924	2969	850	448
	CDH80	12568	22	47	139	470	1468	2295	2479	2519	1925	983	157	64
(14) Wind	WSAvg	7.5	7.5	7.9	8.5	8.5	8.0	7.0	6.2	6.4	7.4	7.9	7.6	7.5
(15) Precipitation	PrecAvg	50.2	2.5	2.7	3.0	2.4	2.5	6.9	8.4	8.6	6.7	2.6	1.5	2.4
	PrecMax	69.6	5.5	11.6	9.1	8.6	8.9	14.1	15.2	13.6	13.6	6.6	4.8	15.6
	PrecMin	38.6	0.4	0.4	0.3	0.2	0.2	3.1	4.0	4.6	1.7	0.4	0.2	0.5
	PrecStd	8.0	1.5	2.3	2.4	1.7	2.1	3.0	2.6	2.8	3.1	1.4	1.1	3.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.4	84.4	86.3	89.4	93.4	94.4	95.0	95.0	93.8	91.3	86.6	84.3
		MCWB	71.3	71.1	70.8	73.3	75.0	78.3	79.4	80.4	79.2	77.3	75.7	73.8
	2%	DB	79.3	81.3	83.7	87.0	90.8	92.8	92.9	93.1	92.2	89.7	84.0	81.8
		MCWB	69.5	69.9	71.2	72.3	74.6	78.3	79.5	80.0	78.7	76.7	73.6	71.8
	5%	DB	76.3	78.7	81.4	84.9	89.2	91.1	91.2	91.4	90.8	87.8	81.6	79.2
		MCWB	67.9	68.7	70.3	71.6	74.1	78.1	79.3	79.7	78.4	75.8	71.4	70.8
	10%	DB	72.7	75.3	78.9	82.5	87.4	89.9	90.0	90.2	89.5	85.8	79.2	76.0
		MCWB	66.5	67.1	68.6	70.7	73.6	77.9	79.2	79.4	78.3	74.9	69.8	69.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.3	74.4	77.0	78.9	79.8	83.1	83.3	84.1	83.3	81.6	80.0	77.4
		MCDB	78.9	79.8	81.1	84.3	86.7	89.3	89.3	90.0	88.4	86.2	83.6	81.3
	2%	WB	71.6	72.6	74.8	76.7	78.3	81.8	82.2	82.8	82.0	79.9	76.2	74.3
		MCDB	76.6	78.0	79.1	81.8	85.5	88.0	88.6	89.4	87.6	85.2	80.3	78.2
	5%	WB	69.8	70.9	73.0	75.0	77.3	80.6	81.3	81.9	80.9	78.5	74.3	72.6
		MCDB	73.9	75.9	77.5	80.5	84.5	87.3	87.9	88.4	86.7	83.9	78.7	76.4
	10%	WB	68.0	69.3	71.2	73.5	76.3	79.6	80.3	80.9	79.7	77.3	72.4	71.0
		MCDB	71.6	73.3	75.9	79.4	83.2	86.7	86.8	87.1	85.6	82.6	76.8	74.6
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	18.5	18.5	18.1	17.6	15.4	14.6	14.5	15.2	17.1	18.5	18.2
		MCDBR	20.3	19.9	19.8	20.1	19.3	17.1	16.1	16.3	17.0	17.6	18.5	18.9
		MCWBR	11.7	10.4	9.6	8.6	7.5	6.5	6.7	6.4	6.7	7.6	9.5	10.1
	5% WB	MCDBR	17.7	16.8	16.1	16.8	15.9	15.6	15.3	15.3	15.6	15.3	15.5	16.6
		MCWBR	12.1	10.4	10.1	9.7	7.3	7.2	7.2	7.2	7.3	7.8	9.6	10.4
(40) Clear-Sky Solar Irradiance	taub		0.342	0.355	0.368	0.392	0.424	0.449	0.473	0.453	0.422	0.390	0.354	0.343
	taud		2.523	2.495	2.445	2.341	2.310	2.321	2.258	2.327	2.382	2.456	2.501	2.548
	Ebn at Noon		286	291	293	287	276	268	261	266	272	276	279	281
	Edh at Noon		29	32	36	40	42	41	44	41	37	33	29	27
(44) All-Sky Solar Radiation	RadAvg		1127	1362	1719	2060	2182	1969	1866	1796	1635	1510	1230	1043
	RadStd		81	87	140	99	119	159	113	113	105	93	63	86

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-5	-81	+311	+218

Nomenclature: See separate page